Concentrations and speciation of polybrominated diphenyl ethers in human amniotic fluid

Polybrominated diphenyl ethers (PBDEs) are persistent organic chemicals used as flame retardants in textiles, plastics, and consumer products. Although PBDE accumulation in humans has been noted since the 1970s, few studies have investigated PBDEs within the gestational compartment, and none to date has identified levels in amniotic fluid. The present study reports congener-specific brominated diphenyl ether (BDE) concentrations in second-trimester clinical amniotic fluid samples collected in 2009 from fifteen women in southeast Michigan, USA. Twenty-one BDE congeners were measured by GC/MS/NCI. The average total PBDE concentration was 3795 pg/ml amniotic fluid (range: 337 – 21842 pg/ml). BDE-47 and BDE-99 were identified in all samples. Based on median concentrations, the dominant congeners were BDE-208, 209, 203, 206, 207, and 47 representing 23, 16, 12, 10, 9 and 6%, respectively, of the total detected PBDEs. PBDE concentrations were identified in all amniotic fluid samples from southeast Michigan, supporting a need for further investigations of fetal exposure pathways and potential impacts on perinatal health.